

SECTION A

1 (**short**) Find all of the possible values of the determinant of the matrix **A** given by

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}$$

where $a^3 = 1$.

[10]

2 (**short**) Sketch carefully the curve $y = \frac{1}{x} - \cos x$ for $-2\pi < x < 2\pi$, marking on your sketch the approximate positions of the two intersections with the x axis that are nearest to $x = 0$.

[10]

3 (**short**)

(a) A real, symmetric 3×3 matrix **A** has eigenvectors $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^t$ and $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^t$ and corresponding eigenvalues 3 and 2. If the third eigenvalue is 1, find such a matrix **A**.

[6]

(b) Describe two simple checks that could help verify that your answer is correct.

[4]